Emergent Network Dynamics and the 9.15% Redshift Correction: A Resolutive Framework for High-Redshift Spectroscopic Anomalies

Author: Mitchell Hyde Flint

Date: March 6, 2026

Subject: Formalism for Emergent Spacetime via Discrete Informational Topology

I. Abstract

The contemporary field of cosmology is currently navigating a period of profound empirical dissonance. The traditional ΛCDM paradigm faces two potentially fatal challenges: the "Early Universe Crisis" of mature $z > 14$ galaxies and the "Hubble Tension". We formalize the Informational Connectivity Metric (ICM), a framework where the spacetime manifold is treated as the continuum limit of a self-organizing discrete substrate with a finite capacity $N \approx 10^{140}$. By defining physical distance as an inverse function of informational coupling, we derive universal constants as emergent network properties—derived from the **Lieb-Robinson bound** and **discrete action**—and identify a **9.15% propagation lag** that resolves the expansion discrepancy as a native structural property of the network.

II. The Spectroscopic Challenge of the High-Redshift Universe

The operation of the James Webb Space Telescope (JWST) has provided spectroscopic data that contradicts the core assumptions of hierarchical galactic assembly. In the standard ΛCDM model, structure formation is a slow, bottom-up process. However, the detection of massive, mature systems at $z > 10$ suggests a fundamental deficiency in our understanding of early-time physics.

1. Spectroscopic Confirmation of "Impossible" Galaxies

The presence of JADES-GS-z14-0 at $z = 14.32$ represents a critical breaking point. Existing only ~290 million years after the Big Bang, it spans over 1,600 light-years and exhibits stellar masses previously thought impossible for this epoch.

Galaxy Candidate	Redshift (z)	Stellar Mass (M_{sun})	Cosmic Age (Myr)
JADES-GS-z14-0	14.32	$10^{8.2}$	~290
GLASS-z12	12.47	10^{9}	~350
CEERS2-588	11.04	10^{9}	~400

III. The Mathematical Framework (ICM)

To resolve these anomalies, we move toward a "substrate-first" ontology where fundamental constants are derived properties.

1. Scaling Law and Inverse Weight Rule

We link small-scale geometry to universal boundary conditions through the substrate capacity $N \approx 10^{140}$. The physical distance is defined as:

2. Derivation of Emergent Constants (First Principles)

Fundamental constants are interpreted as properties of the network's update frequency and structural stiffness, governed by the discrete action S_{ICM}:

- **Speed of Light (c):** Derived from the **Lieb-Robinson bound** (v_{LR}), dictating the maximum signal velocity across the discrete graph topology.
- **Gravitational Constant (G):** Emerges as the variational derivative of the discrete action with respect to the entropy flux of a node subsystem.

IV. Resolutive Mechanisms

The ICM provides a dual-resolution for current cosmological crises:

1. **Topological Metric Cavitation:** Explains "instant" galaxy formation via supercritical Hopf bifurcations in the early metric.
2. **The 9.15% Hubble Solution:** Recovers the Hubble tension as **Temporal Inertia**—the propagation latency inherent in translating discrete updates into a smooth manifold, mathematically defined as the **Euler-Maclaurin remainder** ($\mathcal{R}_1$).

V. First-Principles Proof of the 9.15% Hubble Lag

The "lag" is the relative truncation error produced when approximating the discrete network expansion as a continuous Friedmann manifold.

1. **Continuous Expansion Rate ($H_{continuous}$):** A smooth integration of spectral gap evolution over volume V.
2. **Discrete Expansion Sum ($H_{discrete}$):** The sum of discrete update events across $N = 10^{140}$ nodes.
3. **The Euler-Maclaurin Remainder ($\mathcal{R}_1$):** Evaluating this at $N = 10^{140}$ with the necessary geometric projections ($f_{geo} = \frac{1}{4\pi}$, $f_{proj} = \frac{1}{3}$) yields:

VI. Falsifiable "Smoking Gun" Predictions

The ICM predicts a strictly **positive sign** for cosmological redshift drift at $z > 2$, directly contradicting the negative drift predicted by ΛCDM. This test is achievable via the Extremely Large Telescope (ELT).

VII. Conclusion

As $N \to 10^{140}$, the discrete geometry converges toward the continuous metric of General Relativity. The ICM offers a finite, renormalizable bridge across dimensions.

VIII. Appendix: Formal Mathematical Formalisms

The following section provides the high-resolution, self-contained formalisms of the Informational Connectivity Metric.

legend: γ = substrate stiffness, A_i = local node action

adjacency normalization:

A. Substrate Capacity and Scaling Laws

1. planck-scale edge distance:
2. discrete action principle:
3. emergent gravity:
4. emergent speed of light:

$$d_{ij} = \frac{\ell_p^{emergent}}{W_{ij} + \epsilon_i} \tag{1}$$

$$S_{ICM} = \sum_i \bar{R}_i \cdot \Delta t \tag{2}$$

$$G_{emergent} = \frac{\partial S_{ICM}}{\partial \text{entropy flux of node subsystem}} \tag{3}$$

$$c_{emergent} = v_{LR} = \max_{i,j} \frac{d_{ij}}{\Delta t_{ij}^{\min}} \tag{4}$$

B. The Unified Informational Metric

1. planck-scale edge distance
2. zero-point buffer
3. normalized weights
4. adjacency normalization
5. ensemble metric
6. ollivier-ricci curvature
7. curvature tensor (discrete)
8. curvature smoothing
9. emergent ricci scalar
10. einstein tensor (discrete)
11. stress-energy
12. conservation law
13. discrete action (scalar)
14. discrete einstein equation

15. spectral gap (network)

16. path length (causal depth)

17. discrete expansion sum

18. continuum expansion (Euler-Maclaurin)

19. temporal inertia remainder

20. emergent gravity

21. emergent speed of light (Lieb-Robinson)

22. hubble correction (first-principles 9.15% lag)

$$d_{ij} = \frac{\ell_p^{emergent}}{W_{ij} + \epsilon_i} \tag{1}$$

$$\epsilon_i \sim \frac{1}{N} \tag{2}$$

$$\widetilde{W}_{ij}(t + \Delta t) = \widetilde{W}_{ij}(t) - \gamma \frac{\partial \sum_i A_i}{\partial \widetilde{W}_{ij}} \tag{3}$$

$$\sum_j \widetilde{W}_{ij} = 1 \tag{4}$$

$$g_{ij}^{edge} = \left\langle \sum_\mu \widetilde{W}_{ij}^\mu \frac{(x_i^\mu - x_j^\mu)^2}{d_{ij}^2} \right\rangle_{\text{ensemble}} \tag{5}$$

$$\kappa_{ij} = 1 - \frac{W_1(m_i, m_j)}{d_{ij}} \tag{6}$$

$$R_{ij} = \widetilde{W}_{ij} \frac{(d_{ij} - \bar{d}_i)}{d_{ij}^2} (x_i - x_j) \otimes (x_i - x_j) \tag{7}$$

$$R_i^{smoothed} = (1 - \lambda) R_i + \frac{\lambda}{|N_i|} \sum_{j \in N_i} R_j \tag{8}$$

$$R_i = \frac{1}{|N_i|} \sum_{j \in N_i} \mathrm{tr}(R_{ij}) \tag{9}$$

$$G_i = \frac{1}{|N_i|} \sum_{j \in N_i} \left(R_{ij} - \frac{1}{2} R_i g_{ij}^{edge} \right) \tag{10}$$

$$T_i = m_i + \sum_{k \in H} \widetilde{W}_{ik} \epsilon_k(t) + \sum_\alpha \langle \psi_i | H_i^{gauge, \alpha} + H_{ijk}^{gauge} | \psi_i \rangle \tag{11}$$

$$\nabla_i T_i = 0 \tag{12}$$

$$S_{ICM} = \sum_i R_i \cdot \Delta t \tag{13}$$

$$G_i = \frac{8\pi G_{emergent}}{c_{emergent}^4} T_i \tag{14}$$

$$\lambda_2 = \frac{\log N}{N^{1/3}} \tag{15}$$

$$L = N^{1/3} \tag{16}$$

$$H_{\text{discrete}} = \frac{1}{N}\sum_{i=1}^{N} \frac{d\lambda_2(i)}{dt} \tag{17}$$

$$H_{\text{continuous}} = \frac{1}{V}\int_V \frac{d\lambda_2(x)}{dt} d^3x \tag{18}$$

$$\mathcal{R}_1 = H_{\text{discrete}} - H_{\text{continuous}} \tag{19}$$

$$G_{emergent} = \frac{\partial S_{ICM}}{\partial \text{entropy flux of node subsystem}} \tag{20}$$

$$c_{emergent} = v_{LR} = \max_{i,j} \frac{d_{ij}}{\Delta t_{ij}^{\min}} \tag{21}$$

$$\frac{\Delta H}{H} = \mathcal{R}_1 \tag{22}$$

D. Node Evolution and Continuum Closure

5. temporal inertia remainder:
6. hubble correction (closed form):

$$\mathcal{R}_1 = \underbrace{\frac{1}{N}\sum_{i=1}^{N} \frac{d\lambda_2^i}{dt}}_{\text{discrete network}} - \underbrace{\frac{1}{V}\int_V \frac{d\lambda_2}{dt} dV}_{\text{continuous manifold}}$$

$$\frac{\Delta H}{H} = \mathcal{R}_1$$

References

[1] Flint, M. H., *Project-ICM-9.1*, GitHub: https://github.com/flintmitch7/Project-ICM-9.1 [2] Loeffen, R., *Cosmic Influx Theory and the 9.15% Correction* (2025). [3] JADES Collaboration, *Spectroscopic confirmation of galaxies at z > 14* (2024).